ANUSH SRIRAM RAMESH

Robotics & ADAS Software Engineer | C++ State Estimation & SLAM | Systems Integration & Validation

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SUMMARY

Robotics and ADAS software engineer working on both sides of autonomy: real-time C++ perception and state estimation (multi-camera tracking, camera-IMU-GPS fusion, GTSAM/iSAM2 factor graphs), and the systems integration and validation that take features to production vehicles. Brings 5+ years of real-time automotive software experience from Bosch and Lucid Motors, currently bringing up, integrating and validating ADAS parking features.

EXPERIENCE

Lucid Motors Newark, CA

ADAS VALIDATION ENGINEER Oct 2025 – Present

- Brought up Curb Rash Alert from early integration to production vehicles, connecting perception on the ADAS ECU to the camera server on the infotainment ECU, and validated the integration on the bench against CANoe-simulated system states.
- Built an unsupervised ground-truth annotation pipeline (Python) for dSPACE and AiSIM parking scenarios, used to score perception outputs and widen SOTIF scenario coverage.
- Automated repeatable bench validation of L3 ADAS requirements (CAPL, CANoe, Python), making feature testing **5×** faster and catching rare defects that manual testing missed.
- Root-cause feature failures with perception and feature teams, and present quantified results to Tier-1 software suppliers in weekly reviews.

EMBEDDED TEST ENGINEER II Jul 2024 – Sep 2025

- Designed a unified test-automation architecture for AUTOSAR platform software across **21 ECUs** and **3 vehicle programs**, raising feature test coverage from **20% to 97%** and making it a mandatory regression gate for every release.

RESEARCH ASSISTANT – SLAM SOFTWARE DEVELOPER Sep 2023 – Jun 2024

Northeastern University Boston, MA

Maintaining it as open-source VIO-SLAM since Sep 2026: github.com/sriram-0311/VIO-SLAM

- Developed the fast-tracking front end of a multi-camera GPS-visual-inertial SLAM system in C++ (Toyota Research Institute collaboration), tracking features across cameras and associating them with map points for a pose-only GTSAM/iSAM2 factor-graph back end.
- Implemented global relocalization to recover from tracking loss, with Bag-of-Words place recognition and loop closure to bound map drift; OpenCV KD-trees for geo-spatial keyframe lookup cut relocalization latency.
- Ported the lab codebase to GTSAM 4.3 and modern CMake with bit-identical output, and built a deterministic EuRoC benchmark harness (per-stage timing, ATE/RPE) that took the pipeline from crashing on 9 of 11 sequences to completing **all 11**.
- Profiled and reworked real-time tracking, cutting end-to-end time from 243 to **71 ms/frame** and peak memory from 20 to 4.7 GB.
- Added stereo-inertial fusion (IMU preintegration in iSAM2) with an initializer that needs no static start, reducing trajectory error up to 4.4× on fast-motion sequences (V1_02: **14.2 → 3.2 cm**).
- Integrated self-supervised vision features (I-JEPA, DINOv2) as loop-closure candidates in the C++ loop closer, raising place-recognition precision (AP) from 0.38 to **0.63** over Bag-of-Words.
- Fixed data-association, concurrency and memory bugs: loop-closure RANSAC inlier mis-indexing, factor-graph key loss across iSAM2 resets, a tracking/viewer data race on shared poses, and heap corruption found with AddressSanitizer.

SENIOR EMBEDDED SOFTWARE DEVELOPER Jun 2019 – Mar 2022

Bosch Global Software Technologies Coimbatore, India

- Developed C++ software for Ethernet, Bluetooth and Wi-Fi modules across QNX RTOS, Automotive Android and AUTOSAR for a GM Cadillac EV product-validation platform.
- Engineered real-time network traffic-load emulation with asynchronous multithreading and QNX IPC.

SKILLS

Languages: C++17, Python, C, CAPL

Perception & Tracking: feature tracking, data association (RANSAC, KD-tree nearest neighbor), multi-sensor fusion (camera, IMU, GPS), IMU preintegration, GTSAM factor graphs (iSAM2, smart factors), Eigen, OpenCV, OpenGV, DBoW2, PyTorch

Real-Time C++ & Tooling: Linux, QNX RTOS & IPC, multithreading, latency and memory profiling, AddressSanitizer, gdb, CMake, ROS2, NVIDIA Isaac Sim, evo (ATE/RPE), Git, Jenkins CI

Systems & Validation: ADAS integration and V&V (parking assist), requirements-based testing, release gating, SOTIF, AUTOSAR, CAN/CANoe, dSPACE, AiSIM

PROJECTS

ACTIVE-SLAM FRAMEWORK WITH PROVABLE CORRECTNESS (C++, PYTHON, ROS2) Feb 2024 – Present

- Built GTSAM back ends with QCQP/SDP relaxations that certify globally optimal trajectory estimates, benchmarked on multi-sensor streams with ground truth from NVIDIA Isaac Sim over ROS2 Jazzy.

EDUCATION

Master of Science in **Robotics**, *Northeastern University*, Boston, MA May 2024

GPA: 3.94/4.0 | Courses: Mobile Robotics, Computer Vision, Deep Learning, Reinforcement Learning

Bachelor of Engineering in **Robotics and Automation**, *PSG College of Technology*, Coimbatore, India May 2019

GPA: 8.5/10